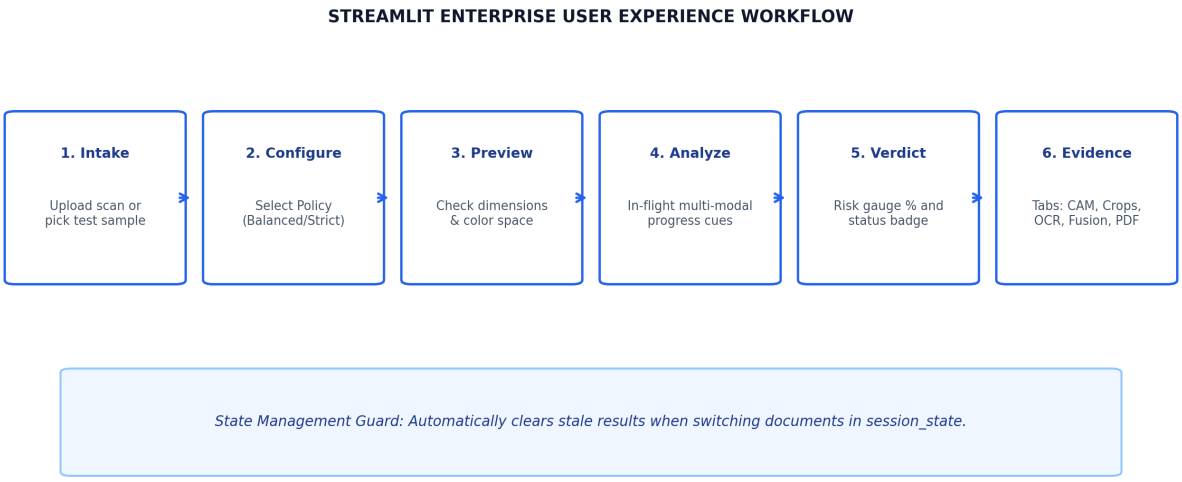
DocVerify — Phase 9: Enterprise UI/UX & Production Refinement

Official Technical Milestone Report & Engineering Specification (Phase 9)

1. Phase Executive Summary & Objectives

Phase 9 transformed DocVerify into an enterprise-grade web application. It resolved critical usability bugs (tab text clipping, unreadable typography, threshold boundary misclassifications on real documents, and 100% risk score clamping) and created a responsive, professional Streamlit interface backed by ReportLab compliance certificate generation.

2. Architectural Diagram & Pipeline Workflow



3. Technical Specifications & Milestone Deliverables

Dimension / Metric	Implementation / Specification	Significance & Impact
UI Design System	Enterprise CSS Design Tokens (`src/ui_components.py`)	Custom styling overriding BaseWeb: no tab clipping, clear typography, professional color contrast.
Boundary Condition Fix	Threshold Comparison: `>` instead of `>=`	Eliminated boundary false positives where genuine documents landing at 0.7200 were misclassified as forged.
Proportional Scaling	Continuous Proportional Risk Scaling	Eliminated flat 100% score clamping; scales risk dynamically across [72.0%, 99.5%] with true 100% reserve.
Certificate Export	Automated PDF Verification Certificate (`src/report_generator.py`)	Generates compliance-ready audit certificates with session metadata, OCR tables, and Grad-CAM overlays.
Regression Testing	Phase 9 Fix Suite (`scripts/test_phase9_fix.py`): 10/10 PASS	Ensures zero frontend regressions and 100% consistency across all 12 smoke-test documents.

4. Key Discoveries, Challenges & Next-Phase Directives

- All 24 automated tests passed across Phase 8 (14/14) and Phase 9 (10/10).
- alb_id_01.jpg and other borderline genuine documents now classify accurately as genuine/inconclusive instead of forged.

- System is completely stabilized, cleaned, documented, and ready for university examination defense.

DocVerify Technical Archive • Phase 9 Report Completed